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Chapter-3 (Isha)

Numerical Method

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Equivalent Values:

Decimal	Binary	Hexa-Decimal
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

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Example 3.1: $(1101.1101)_2 = (?)_{10}$

Integral Part:

$= (\underset{3\,2\,1\,0}{1101})_2$

$= 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 8 + 4 + 0 + 1$

$= 13$

fractional Part:

$= (.\underset{1\,2\,3\,4}{1101})_2$

$= 1\times2^{-1} + 1\times2^{-2} + 0\times2^{-3} + 1\times2^{-4}$

$= 0.5 + 0.25 + 0 + 0.0625$

$= 0.8125$

$\therefore (1101.1101)_2 = (13.8125)_{10}$

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Ex 3.2 Convert the hexa-decimal number 12AF to a decimal number.

Soln: $(12AF)_{16} = (?)_{10}$

$\underset{3\ 2\ 1\ 0}{(12AF)_{16}} = (?)_{10}$

$= 1 \times 16^3 + 2 \times 16^2 + A \times 16^1 + F \times 16^0$

$= 1 \times 16^3 + 2 \times 16^2 + 10 \times 16^1 + 15 \times 16^0$

$= (4783)_{10}$.

Ans.

Extra: $(4561)_8 = (?)_{10}$

$= 4 \times 8^3 + 5 \times 8^2 + 6 \times 8^1 \times 1 \times 8^0$

$= 2048 + 320 + 48 + 1$

$= (2417)_{10}$.

Ans.

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(3.3) ⇒ Convert the decimal 43.375 into its binary equivalent.

⇒ Integral Part:

2 | 43	
2 | 21	−1
2 | 10	−1
2 | 5	−0
2 | 2	−1
	1 −0

$(43)_{10} = (101011)_2$

Fractional part:

Multiplication	Product	Integral
0.375 × 2	0.75	0
0.75 × 2	1.50	1
0.50 × 2	1.00	1

$\therefore (0.375)_{10} = (0.011)_2$

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$\therefore (43.375)_{10} = (101011.011)_2$

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3.4 Convert the decimal number 163 to an octal number.

Soln:

$$8 \mid 163$$
$$8 \mid 20 \quad 3$$
$$2 \quad 4$$

$\therefore (163)_{10} = (243)_8$

3.5 Convert the decimal number 0.65 to its binary equivalent.

⇒

Multiplication	product	Integral part
0.65×2	1.3	1
0.3×2	0.6	0
0.6×2	1.2	1
0.2×2	0.4	0
0.4×2	0.8	0
0.8×2	1.6	1
0.6×2	1.2	1
0.2×2	0.4	0
0.4×2	0.8	0
0.8×2	1.6	1

$\therefore (0.65)_{10} = (.1010011001)_2$.

Ans.

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Example: (3.6) convert the octal number to a hexadecimal number.

$(243)_8 = (?)_{16}$

4 2 1
2 0
2 2 1
3

2 → 010 | 4 → 100 | 3 → 011

Binary Equivalent: 010 100 011

Binary quadruplets: 0000 | 1010 | 0011

8 4 2 1 → 0000 → 0

8 4 2 1 → 1010 → 9 → A

8 4 2 1 → 0011 → 3 → 3

$\therefore (243)_8 = A3_{16}$

-Hexadecimal এ right থেকে 4 করে।

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Example 3.7:

Convert the hexadecimal number 39.B8 to an octal number.

$(39.B8)_{16} = (?)_8$.

⇒ Hexadecimal → 3 9 . B 8

↓ ↓ ↓ ↓

Binary equivalent → 0011 1001 1011 1000 (P)

Binary triplets

000	111	001	101	110	000
↓	↓	↓	↓	↓	↓
0	7	1	5	6	0

(4 2 1 above each triplet)

∴ $(39.B8)_{16} = (71.56)_8$.

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Ex 3.10

0.00596 is expressed as 0.596×10^{-2}

65.7452 " " 0.657452×10^{2}

-486.8 " " -0.4868×10^{3}.

Computer Arithmetic

* floating point arithmetic:

Addition Algorithm:

Let two numbers x and y, and result z.
Let fractional part fx, fy and fz.
and exponent Ex, Ey and Ez.

Algorithm:

1. Set Ez = the larger (বড়টা) of Ex and Ey
(Assume Ex >= Ey then Ez = Ex

2. Shift right - shift fy to the right by
Ex - Ey places.

3. Add Set fz = fx + fy.

4. Normalise - If the absolute value of fz is greater than one, shift decimal point of fz to the left of the

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significant digit and then increase by One then $z = f_z \times 10^{E_z}$

Ex 3.13: Add the numbers 0.964572 E2 and 0.586351 E5

$\Rightarrow E_z = 5$

$f_x = 0.964572\ E2$

$= 0.000964\ E5$

$f_y = 0.586351\ E5$

$f_z = f_x + f_y$

$= 0.000964 + 0.586351$

$= 0.587315$

$\therefore z = 0.587315 \times 10^5$

$= 0.587315\ E5$

- right shift করলে decrease
- left হলে (+) increment হয়

0. এর পর ২টা 0

Ex - 3.14 Add the numbers 0.735816 E4 and 0.635742 E(4)=7

$\Rightarrow E_z = 4$

$f_x = 0.735816$

$f_y = 0.635742$

$f_z = f_x + f_y$ — Normalise করতে হবে

$= 1.371558\ E4$

$= 0.137155\ E5$

$\Rightarrow Ez = 5$

$x = 0.964572\ E2$

$y = 0.586351\ E5$

$x = 0.964572\ E2$ — E3 & E5

$= \dfrac{0.964572 \times 1000}{1000}\ E2$

$= 0.000096457 2\ E5$

$f(x) = 0.000096457 2\ E5, f(y) = 0.586351 E5$

$f(z) = f(x) + f(y) = 0.587315572.$

$\therefore Z = 0.587315572\ E5$

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Substraction:

Ex3.15 Substract the number 0.994576E-3 From 0.999658E-3.

$\Rightarrow E_z = -3$

$f_y = 0.994576$

$f_x = 0.999658$

$f_z = f_x - f_y = 0.005082\ E-3$

$= 0.508200\ E-5.$

normalise

$0.123\ldots$

Ans.

(A) Subtract the numbers 0.471255 EX and 0.853196 EG.

$\Rightarrow E_z = EX$

$f_x = 0.471255\ EX$

$f_y = 0.853196$

$= 0.085319$

0.853196

0.08

0.471255

$f_z = \cancel{0.853196} - 0.085319$

$= 0.385936$

$Z = 0.385936\ EX.$

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Ex 3.16: Multiply the numbers 0.200000 E1 and 0.400000 E-2.

$\Rightarrow$ $F_z = f_x \times f_y$

$= 0.2000000 \times 0.400000$

$= 0.0800000$

$E_z = E_x + E_y$

$= 4 + (-2)$

$= 2$

$Z = 0.0800000\ E2$

(R)

$Z = 0.800000\ E1$

Ans.

Ex 3.17: Divide the number 0.876543 E-5 by 0.200000 E-3.

$\Rightarrow f_z = 0.876543 \div 0.200000$

$= 4.382715$

$E_z = (-5) - (-3)$

$= -5 + 3$

$= -2$

left এগোলে Exponent (+)হয়

$-2+1$

$= -1$

$\therefore z = 4.382715\ E-2$

left ↑ ↓ normalise

$0.438271\ E-1$ (normalise)

Ans

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Extra: Add the numbers 0.735816 E4 & 0.635742 E-2

$\Rightarrow Ez = 4$

$x = 0.735816\ E4$

$y = 0.635742\ E-2$

$\therefore y = \dfrac{0.635742 \times 10^{-2}}{10^6}$

$= 0.000000\ 635\ E4$

$= 0.000000\ E4$

$z = fx + fy$

$= 0.735816 + 0.000000$

$= 0.735816$

* Add the numbers 0.500000 E1 and 0.100000E-7.

$\Rightarrow Ez = E1$

$x = 0.500000\ E1$

$y = 0.100000E-7$ E1

$= \dfrac{0.100000 \times 10^8}{100000000}$ E-7

$= 0.00000000\ 1$ E1

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$f(x) = 0.500000$

$f(y) = 0.000000001$

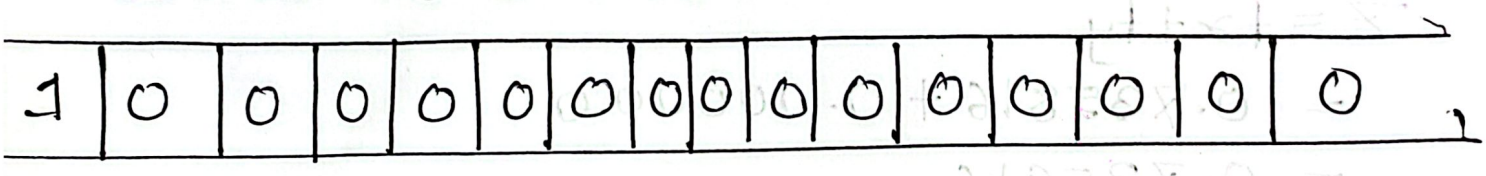
$f(z) = f(x) + f(y)$

$= 0.500000001$

$z = 0.500000001E1$

∴ Ans.

Ex 3.9 (Book) Show that the number -32768 is represented in 16 bit word as follows:



⇒ Binary digit 32768 is

1000000000000000

1's comp → 0111111111111111

2's " → +1

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2	32768
2	16384 - 0
2	8192 - 0
2	4096 - 0
2	2048 - 0
2	1024 - 0
2	512 - 0
2	256 - 0
2	128 - 0
2	64 - 0
2	32 - 0
2	16 - 0
2	8 - 0
2	4 - 0
2	2 - 0
2	1 - 0
	0 - 1

Binary digit of 32768 is 1000000000000000

1's comp → 0111111111111111

2's " → 1

Sign bit 1 0000000000000000

∴ −32768 = 1000000000000000

Ans.

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Example 4.2: (Book)

What is the accuracy of the following number?

1. 95.763 → significant number → 5
2. 0.008472 → 4 → 4 → 4
   (1 leading zero)
3. 0.045 6000 → 4 → 4 → 6
4. 36 → 2 → 2 → 2

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5) 3600 → significant number → 2

6) 3600.00 → 4 4 → 6

Precision: যার decimal বেশি তার Precision বেশি। $\boxed{\text{Precision} = 10^{-n}}$

Ex4.1 (Book)

a) 4.3201 ⇒

⇒ Number of decimal Places = 4

∴ Precision = 10^{-4} ∴ $4.3201 = 43201 \times 10^{-4}$

b) 4.32

Number of decimal Places = 2

∴ Precision = 10^{-2} ∴ $4.32 = 432 \times 10^{-2}$

c) 4.320106

Number of decimal Places = 6

∴ Precision = 10^{-6} ∴ $4.320106 = 4320106 \times 1$

∴ 10^{-6} is the smallest value, it has th. greatest Precision.

∴ 4.320106 has the great.

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Chapter 6

(Roots of Nonlinear Equations)

* Algebric equation $\Rightarrow y = f(x) \rightarrow f(x, y) = 0$

  Ex $\Rightarrow$ $3x + 4y = 0$

* Polynomial $\Rightarrow$ $ax^2 + bx + c = 0$;

  $x^3 - 4x + 4 = 0$

* Transcendental $\rightarrow$ $\sin x$, $\cos x$, e^x, $\log x$, etc.

Algebric :

$3x + 5y - 21 = 0$ (Linear)

$2x + 3xy - 25 = 0$ (non-linear)

$x^2 - 2xy + y^2 = 0$ (")

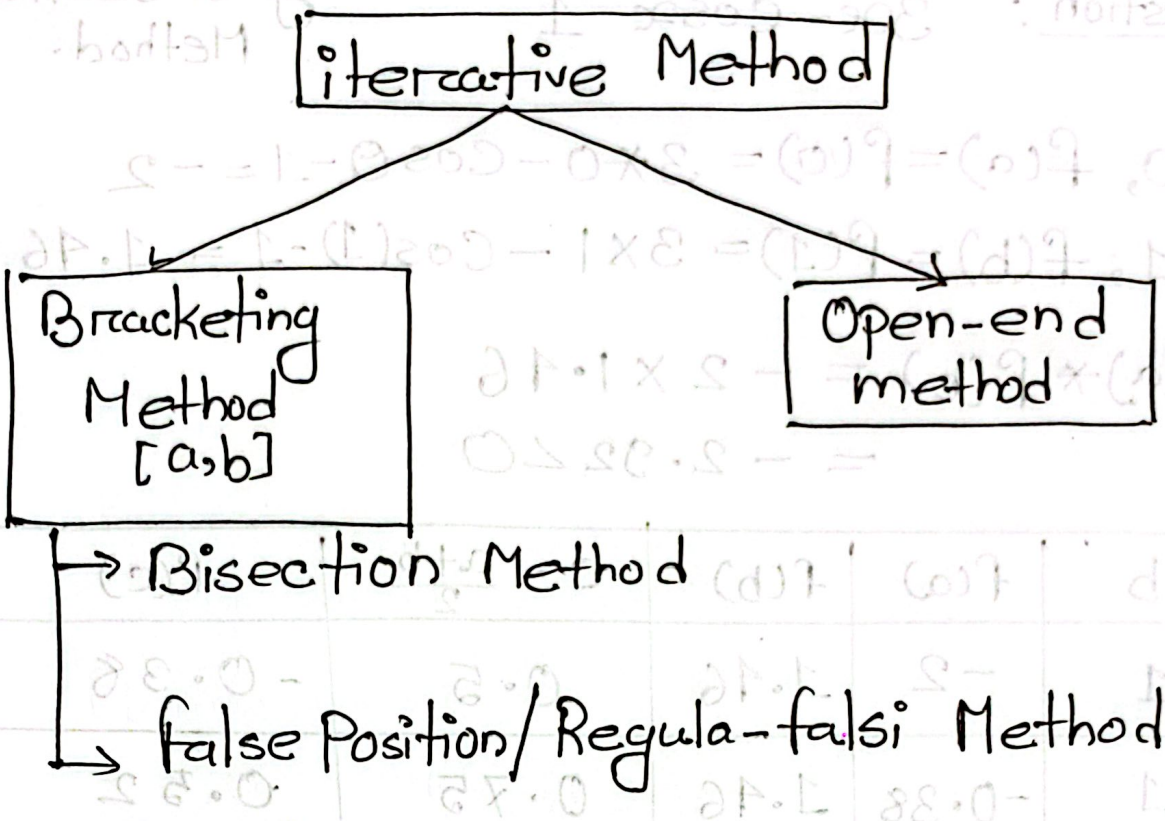
Roots/Zero's:

The values of x which satisfy the equation $f(x) = 0$ are called roots of the equation / the zeros of $f(x)$.

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Algorithm for Bi-section Method:-

1. Choose two real numbers a & b. such that $f(a) * f(b) < 0$.
2. define root, $c = \frac{a+b}{2}$
3. find $f(c)$.
4. if $f(a) * f(c) \leq 0$ then set $b = c$ else $a = c$

Return to step 1 until finding the root matched twice.

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Find the root from the following function using bi-section Method.

Question: $3x - \cos x - 1$

$\Rightarrow a = 0,\ f(a) = f(0) = 3*0 - \cos 0 - 1 = -2$

$b = 1;\ f(b) = f(1) = 3\times 1 - \cos(1) - 1 = 1.46$

$\therefore f(a) * f(b) = -2 \times 1.46$

$= -2.92 < 0$

a	b	f(a)	f(b)	$c = \frac{a+b}{2}$	f(c)
0	1	-2	1.46	0.5	-0.38
0.5	1	-0.38	1.46	0.75	0.52
0.5	0.75	-0.38	0.52	0.625	0.06
0.5	0.625	-0.38	0.06	0.56	-0.16
0.56	0.625	-0.16	0.06	0.59	-0.05
0.59	0.625	-0.05	0.06	0.61 ✓	0.00
0.59	0.61	0.05	0.00	0.60	-0.02
0.60	0.61	-0.02	0.00	0.61 ✓	-0.01

Root = 0.61

$f(a) * f(c) \rightarrow$ Positive

$a = c$

বা $b = c$.

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